Intracity Temperature Estimation by Physics Informed Neural Network Using Modeled Forcing Meteorology and Multispectral Satellite Imagery

Donghang Wu, Weiquan Liu, *Member, IEEE*, Bowen Fang, Linwei Chen, Yu Zang, *Member, IEEE*, Lei Zhao, Shenlong Wang, *Member, IEEE*, Cheng Wang, *Senior Member, IEEE*, José Marcato, Jr., *Member, IEEE*, and Jonathan Li, *Senior Member, IEEE*

***Abstract*— Estimating urban surface temperature at high resolution is crucial for effective urban planning for climate-driven risks. This high-resolution surface temperature over broader scales can usually be obtained via satellite remote sensing for historical period. However, it can be hard for future predictions. This article presents a physics informed hierarchical perception (PIHP) network, a novel approach for accurate, high-resolution, and generalizable urban surface temperature estimation. The key to our approach is leveraging the implied temperature-related physics information of the land surface structure from high-resolution multispectral satellite images, thus achieving precise estimation or prediction for high spatial resolution urban surface temperature. Specifically, a semantic category histogram is first designed to describe the land surface structures. Based on this, a hierarchical urban surface perception network is proposed to capture the complex relationship between the underlying land surface features, upper atmosphere conditions, and the intracity temperature. The proposed PIHP-Net makes it possible to generate models that can generalize across different cities, thus estimating or predicting high-resolution urban surface temperature when the satellite land surface temperature (LST) observation is not available. Experiments over various cities in different climate regions in China show, for the first time, errors less than 2 K (for most of the cases) at the high resolution (60-by-60 meters grids), thus making it possible to predict future *intracity temperature* from forcing meteorology and multispectral satellite imagery.**

***Index Terms*— Deep neural network, downscaling, land surface temperature (LST), multispectral satellite imagery.**

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Donghang Wu, Weiquan Liu, Linwei Chen, Yu Zang, and Cheng Wang are with the Fujian Key Laboratory of Sensing and Computing for Smart Cities, School of Informatics, Xiamen University, Xiamen 361005, China (e-mail: donghangwu@stu.xmu.edu.cn; wqliu@xmu.edu.cn; willim@stu.xmu.edu.cn; zangyu7@126.com; cwang@xmu.edu.cn).

Bowen Fang and Lei Zhao are with the Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign, Urbana, IL 61801 USA (e-mail: bowenf2@illinois.edu; leizhao.yale@gmail.com).

Shenlong Wang is with the Department of Computer Science, University of Illinois at Urbana-Champaign, Urbana, IL 61801 USA (e-mail: shenlong@illinois.edu).

José Marcato, Jr., is with the Geomatics Laboratory, Faculty of Engineering, Architecture and Urbanism and Geography, Federal University of Mato Grosso do Sul, Campo Grande 79070, Brazil (e-mail: jose.marcato@ufms.br).

Jonathan Li is with the GeoSTARS Laboratory, Department of Geography and Environmental Management, University of Waterloo, Waterloo, ON N2L 3G1, Canada (e-mail: junli@uwaterloo.ca).

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I. Introduction

CITIES, as the hotspots of concentrated population and infrastructure, are where major climate-driven impacts occur [1], [2]. Effective urban planning and infrastructure-based growth strategies rely on high-resolution urban climate predictions [3], [4]. High-resolution and high-precision land surface temperature (LST) prediction has been extremely challenging, especially over urban surfaces, because of their large heterogeneity [5], or the complicated natural and human behaviors, such as seasonal change, diurnal temperature differences, urbanization, population density, energy structure, etc. [6], [7], [8], [9]. We acknowledge that "high resolution" has different definitions for different fields. For example, subhundred meters resolution is typical in satellite remote sensing retrieval studies, and might not be considered as high resolution. However, in climate modeling studies, predicting intracity LST for future time horizon could be very challenging. Seasonal forecasts or long-term projections from climate models are either at coarse resolutions (>1 km) [3], [4] or completely missing urban landscapes [10]. The objective of this study is to establish a machine learning-based framework to predict intracity LST that can be used for urban climate applications using remote sensing data and deep neural networks, rather than LST retrieval. Therefore, the definition of "high resolution" in this study means subhundred meters, aligned with urban climate studies.

Recent works [10], [11] employed the physics informed machine learning (PIML) paradigm to build an urban climate emulator to predict the citywide average temperatures on the global scale, demonstrating the large potential of incorporating

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physical understanding into machine learning for the urban temperature prediction. These PIML methods have also been demonstrated in modeling weather and climate processes [12] and other ecosystems [13]. Such a trend inspires us to deeply dig into the underlying physical reasons for the temperature prediction, not just considering it a sheer computer vision problem.

State-of-the-art methods to estimate LST can be mainly classified into two categories: physics-based methods and statistics-based methods. Physics-based models exploit the dynamic processes between the urban canopy and the atmospheric boundary layer to solve the temperature field [14], [15]. These methods are very computationally demanding and thus usually low-resolution (several kilometers and above). On the other hand, statistical methods seek to establish empirical relationships between the LST and the other observational data, such as meteorological variables, land cover, geography, and vegetation indices [16], [17], [18], [19], [20], [21]. However, such methods are limited by the availability of the data and suffer from generalization ability and low accuracy [22].

Dynamic models leverage process-based equations that aim to resolve the physics within the urban canopy and atmospheric boundary layer to solve the temperature field. These models are highly computationally expensive and thus can hardly operate at a high spatial resolution, and have been limited by the physics represented in the model and the availability of urban surface characteristic datasets. For example, the dynamic downscaling methods using some widely used models such as the weather research and forecast (WRF) model [23] are usually conducted at 1–2 km resolution at the finest [4], [24], [25]; whereas the computational fluid dynamics (CFDs)-based models are limited to very small scales such as a single urban block or street canyon [26], [27]. Their modeling accuracy is further subjected to the accuracy of the parameterization and representation of the physical processes in the models. These methods can hardly be applied for high-resolution urban temperature estimation, because of the infeasibility in resolving the very small physics scale.

Statistical models, on the other hand, seek to establish empirical relationships between LST and the auxiliary data, such as land cover, vegetation indices, and/or other observational data [28], [29], [30], [31]. Empirical at its core, traditional statistical downscaling methods are limited by: 1) the complexity of statistical methods used; 2) the availability and reliability of the observed records; 3) the relatively arbitrary choices of the features; and 4) omission of the physics represented in the statistical models [32], [33], [34], [35]. These barriers significantly limit the traditional urban temperature estimation from generalization both spatially (i.e., upscaled to a larger region or applied to other study locations) and temporally (i.e., future forecast).

Recent efforts have started to explore the applications of both deep neural networks and physics-informed machine learning to tackle Earth and environmental science challenges. These applications pointed to some potentially promising avenues to address the aforementioned critical yet unresolved research gaps. Specifically, [36] modeled the lake temperatures across the depth and over time by combining physics-based models and deep learning methods. [37], [38] leveraged the advantages of convolutional neural network (CNN) in the processing of multichannel images [2], [39], [40], applying CNN to sea surface temperatures maps and the oceanic heat content maps. These methods mainly targeted the LST prediction of homogeneous surfaces such as woodlands and waters, but nevertheless demonstrate a promising potential of the physics-informed neural networks paradigm. However, it is unknown whether this paradigm can successfully predict the intracity LST over urban surfaces which are largely heterogenous and with complex 3-D structures.



Fig. 1. Distribution of the selected urban in China.

In this study, we propose a physics informed hierarchical perception network (PIHP-Net) to predict high-resolution urban LST directly from climate modeled forcing meteorology at a higher atmospheric level and land surface satellite imagery. Guided by the process-based physics understanding, such a network leverages the high-resolution multispectral satellite images, which are low cost to acquire, to achieve accurate LST prediction at an high spatial resolution. Specifically, the semantic category histogram is first designed to describe the urban surface structure; then, the LST is decomposed by the bidimensional empirical mode decomposition (BEMD) to capture features at various scales. Based on this, a hierarchical urban surface perception scheme is proposed via a multibranch network structure. This scheme captures the complex relationship between the land surface structures, upper atmosphere conditions and the intracity temperatures, and thus capable of generating accurate urban LST estimation results at a high resolution, when in the situation that LST cannot be generated in the future period or due to cloud cover.

II. Study Area and Data

In this work, 31 major or provincial capital cities in China are selected as the study area, see Fig. 1. The selected cities cover seven administrative regions of China and contain diverse types of terrain including metropolis, mountain, water, etc. We collected a suite of datasets from various sources in the study region to implement our PIHP-Net, described below. All of them are highly urbanized and populated. Research on these cities has great significance for assisting urban planning, carbon neutrality, and climate prediction. Specifically, four

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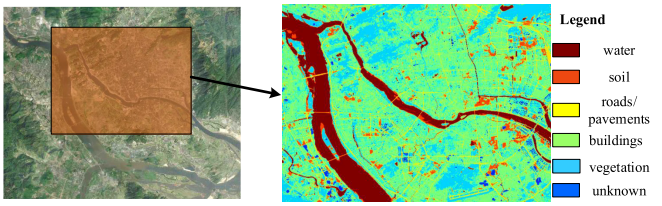


Fig. 2. Illustration of the generation process of GSCFM. (a) Satellite remote sensing images of Fuzhou, China. (b) Visualized map of the labeled. Inferred semantic categories (water, buildings, vegetation, soil, roads/pavements, and unknown) with text colored by the label color.

datasets are collected and processed in this article, as described below:

The first dataset is the Landsat-based ready-to-use (RTU) LST product from the CASEarth DataBank system,[1] which is based on the United States Geological Survey (USGS) Landsat8 Operational Land Imager (OLI)/Thermal Infrared Sensor (TIRS) sensor with a resolution of 30 m. A single-channel algorithm was used to retrieve LST in this product [41]. In this study, we collected the data from 2014 to 2019. All of the selected regions cover the urban area. Since we aim to estimate the block level urban temperature from the satellited image, thus the 60 m-by-60 m of the urban grid has been divided in our experiments. Thus, the corresponding LST data have also been aggregated to 60 m resolution. An example visualized map is shown in Fig. 5 (a).

The second dataset is the Nationwide remote sensing images of the major and provincial capital cities in China, along with the semantic label. Fig. 2 (a) and (b) show the examples of second dataset of Fuzhou, China, in which (b) is the visualized result of the semantic label, with the different urban surface properties denoting as various colors. These multispectral remote sensing images have a resolution of 1 m from the Google Earth website[2] for all major provinces and key cities in China. Besides the spectral information, we find that the LST is also highly correlated with the properties of the underlying surface. So, a pixelwised labeling of the remote sensing image is also involved in this dataset. Specifically, each pixel is automatically labeled with predefined categories (including water, buildings, vegetation, soil, and roads/pavements, etc.) by the state-of-the-art classification technique [42] and the accuracy of the classification is 96.5%.

The third dataset is the temperature measured a by the weather station from the website of China's National Greenhouse Data System.[3] The dataset contains temperatures measured at weather stations in urban centers in 30 cities for each day from 2014 to 2019. And the kind of temperatures used is daily average air temperature at 2 m.

The fourth dataset is the normalized difference vegetation index (NDVI) of the Landsat-based ready to use (RTU) products from the CASEarth DataBank system,[4] which is based on the USGS's Landsat8 OLI/TIRS sensor with a resolution of 30 m. And normalized difference built-up index (NDBI), for Landsat 8 data from USGS global visualization viewer website,[5] NDBI = (Band 6 − Band 5)/(Band 6 + Band5) [43]. Meanwhile, all of the selected regions are the same to first dataset.

The last dataset is the atmospheric forcing data, provided by the National Aeronautics and Space Administration (NASA) Modern Era Retrospective-Analysis for Research (MERRA)-2 reanalysis data system [44]. These data are publicly accessible from the NASA MERRA-2 website.[6] The resolution of such data is 0.5° latitude × 0.625° longitude. The atmospheric forcing data have a temporal resolution of days and reflect the overall weather conditions of the city on that day. A list of these variables is shown in Table I.

TABLE I

ATMOSPHERIC AND LOCATION DATA. THE AIR TEMPERATURE IN THIS STUDY MEANS THE ATMOSPHERIC TEMPERATURE AT THE REFERENCE HEIGHT (60m ABOVE THE SURFACE CANOPY TOP) IN REANALYSIS DATA OR CLIMATE MODELS

Type	Name
Land Surface Forcings	Surface absorbed longwave radiation
	Surface income shortwave flux
Land Surface Diagnostics	Total precipitation land
Single-Level Diagnostics	Atmospheric temperature max
	Atmospheric temperature mean
	Atmospheric temperature min
Analyzed Meteorological Fields	Surface pressure
	Atmospheric temperature at the reference height
	Eastward wind
	Northward wind
	Specific humidity

Due to the limitation of national policy, the raw high-resolution multispectral satellite remote sensing images are not open source. As an alternative, we open-source the calculated [geographical semantic category fraction matrix (GSCFM), details can be viewed in Section III] of all cities. The complete dataset will be released via the File Transfer Protocol (FTP) server later.

III. METHODS

Considering the temperature prediction as a simple image-based regression problem, the network may be confused by the complex image pattern. Thus, the relationship between the underlying surface, upper air condition, and the LST may fail to be captured by the network.

We propose a physics-informed hierarchical urban surface temperature perception scheme to guide the overall estimation process by related physical factors. First, by employing previous image ground targets classification network [42], the satellite imagery is first assigned a pixel-wise semantic label according to the geophysical categories. In this work, 11 urban surface categories (cultivated land, garden land, forest land, grassland, buildings, roads, structures, excavated land, bare land, water, unknown, or covered with snow) are applied. Finally, the 11 categories are divided into six major categories. Second, we design a grid-wised (60-by-60 meters) semantic category histogram to further aggregate the urban surface features. Such a dense urban surface descriptor, and the atmospheric forcing and weather station observation data are fed into the network. Then, by passing a proposed

[1]http://databank.casearth.cn/
[2]https://earth.google.com/
[3]http://data.sheshiyuanyi.com/WeatherData/
[4]http://databank.casearth.cn/
[5]https://glovis.usgs.gov/
[6]https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/

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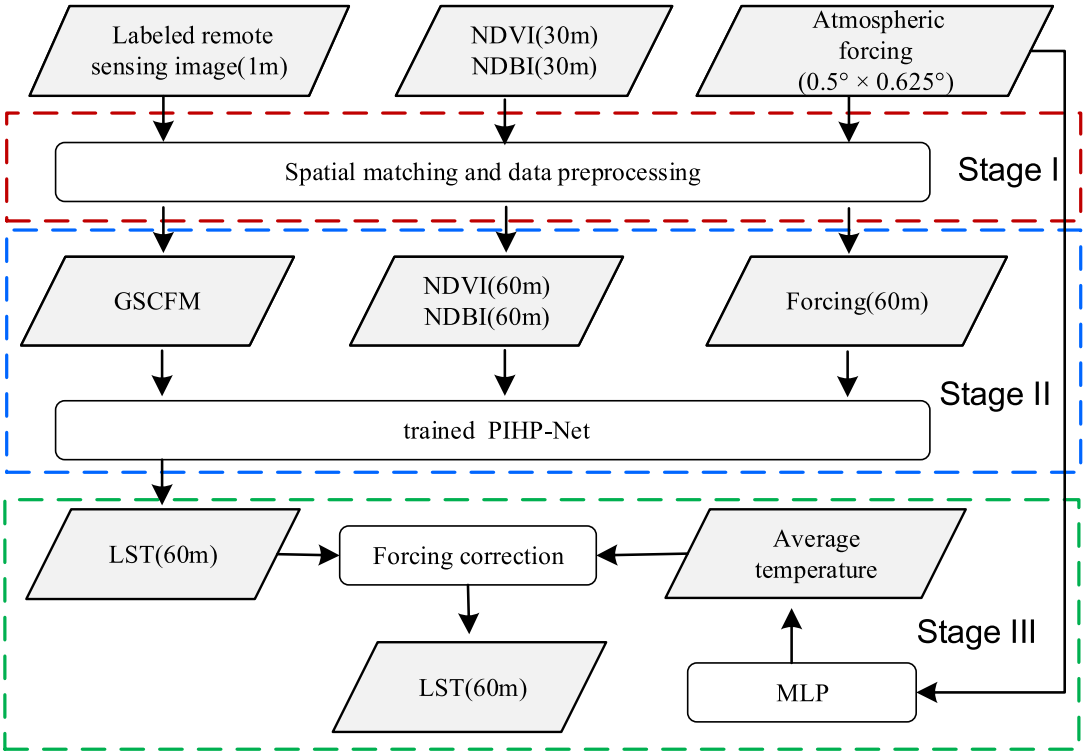


Fig. 3. Flowchart of the proposed method for predicting the 60-m LST from labeled remote sensing image, NDVI, NDBI, and atmospheric forcing data.

BEMD-based hierarchical network, the mapping relationship between the urban surface temperature and the input data can be captured at different scales, producing the estimated temperature for each grid of the testing city.

A flowchart of predicted LST is shown in Fig. 3. Implementation of the whole is performed by three stages: 1) data spatial matching and preprocessing; 2) prediction of LST; and 3) correction and generation of LST. In the stage of data spatial matching and preprocessing, labeled remote sensing image, NDVI, NDBI, and atmospheric forcing are matched under the same space. For NDVI, NDBI, and atmospheric forcing are mapped onto each 60 m grid, different scales of GSCFM are constructed as input to PIHP-Net. In the second stage, LST is predicted through trained PIHP-Net. Finally, in the stage of correction and generation of LST, the LST from stage II goes through forcing correction to obtain a 60 m precision LST. Details of forcing correction are given in Section III-B3.

A. Physics Informed Hierarchical Perception Scheme

1) Geographical Semantic Category Fraction Matrix: To introduce the structural distribution information of local city embedded in remotely sensed image into neural network, a straightforward approach is to import the entire image directly into the network. However, we found that this approach has difficulty capturing the general relationship between temperature and land surface information, because the high complexity of the urban surface features leads to poor generalization of the whole system. Thus, we designed a semantic category histogram to summarize the potential factors that influence the local-scale urban surface temperature. This descriptor is subsequently denoted as the GSCFM.

The GSCFM contains the ground structure information from labeled remote sensing images with a spatial resolution of 1-by-1 m. The ground structure information is set to 5-by-5 m as one cell and calculated from the labeled images within each cell, as described below. Each cell is described by a high-dimensional feature, which implies the underlying surface characteristics that affect the local temperature. In our approach, the five main temperature-related structure categories (including water, buildings, vegetation, soil, and roads/pavements) are summarized by previous semantic labeling works, with various colors in the dots indicating different urban surface properties. The urban LST data were obtained from NASA's Landsat satellite measurements with a spatial resolution of 60-by-60 m. For each 60-by-60 m grid, we build a larger matrix by a batch of 5-by-5 meter cells. The matrix size is $M \times M \times d$, where M denotes the number of cells and d denotes the dimension of the feature vector of one cell. Each category of GSCFM in one cell is defined by the following equation:

$$d(l) = \frac{C(S_l)}{C(S)} \tag{1}$$

where S denotes the set of pixels on a particular cell, $C(\cdot)$ denotes the number of points in S, and S_l denotes the number of pixels with category l in S. Fig. 4 shows examples of the constructed GSCFM.

2) Multiscale Hierarchical Urban Surface Perception: Due to the complexity of the surface structure and the fact that the temperature of a single 60-by-60 m grid may be influenced by the surrounding surface structure, the network is hard to capture the relationship between the GSCFM and LST directly.

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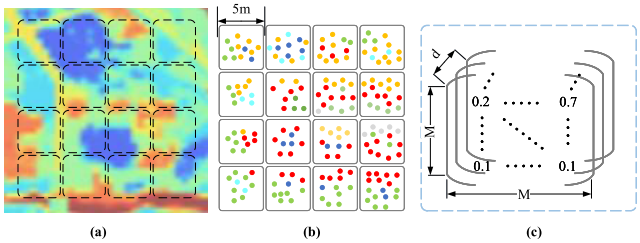


Fig. 4. Illustration of the generation process of GSCFM. (a) Background is part of the land surface semantic label image, and a dummy black box represents the 5-by-5 meter cells. (b) Visualized result of different categories in every cell, different colors represent different categories. (c) For each cell, we can calculate the percentage of each category in one cell. There are d categories in total, so each cell corresponds to a vector of $1 \times d$. And for a 60-by-60 m grid, we can select $M \times M$ cells to describe the surface structure and domain information of this gird, and this is one example of the GSCFM with size $M \times M \times d$.

Inspired by previous signal processing works [45], [46], [47] and to reduce the complexity of the nonlinear transformation in the neural network, we propose a BEMD to decompose raw data into multiscales. Empirical mode decomposition (EMD) is a classical signal analysis tool [48] that decomposes data into a series of low-frequency bases, which are called intrinsic mode functions (IMFs). The BEMD algorithm treats the image as a signal and decomposes it to obtain different IMFs, and each IMF contains the information of different scales. Thus, our idea is to build the connections between the IMFs of the LST image and the GSCFM at different scales. Such a multiscale perception scheme allows the network to capture the relationship between the urban surface and the temperature in various scales, leading to better estimation results.

Consider the LST data as a one-channel image, denoted as $I(\mathbf{p})$. Then, the decomposition process can be formally written as

$$I(\mathbf{p}) = h(\mathbf{p}) + r(\mathbf{p}). \tag{2}$$

Here, $h(\cdot)$ is the intrinsic mode representing fine-scale distribution information. $r(\cdot)$ is the residue layer obtained by averaging the envelopes of local maxima $E(\cdot)$ and minima $e(\cdot)$, i.e., $r(\mathbf{p}) = (E(\mathbf{p}) + e(\mathbf{p}))/2$. For the computation and localization of local extrema, we use directly finding the extrema from each subsquare neighborhood. In addition, we simply use uniform cubic spline interpolation to compute the envelopes $E(\cdot)$ of maxima and $e(\cdot)$ of minima from the local extrema. Such a decomposition can be recursively applied to obtain multiscale IMFs, as written by

$$I(\mathbf{p}) = \sum_{i=1}^{n} h_i(\mathbf{p}) + r_n(\mathbf{p}). \tag{3}$$

Here, n denotes the number of recursions. As shown in Fig. 5, three decomposition operations are applied, where Fig. 5(a) is the original LST image. Fig. 5(b)–(d) are the acquired IMFs that reflect the temperature signals in different scales. Fig. 5(e) is the residual image after three iterations. These modes along with the residual are connected with different branches of the network, to guide the GSCFM in different spatial scales.

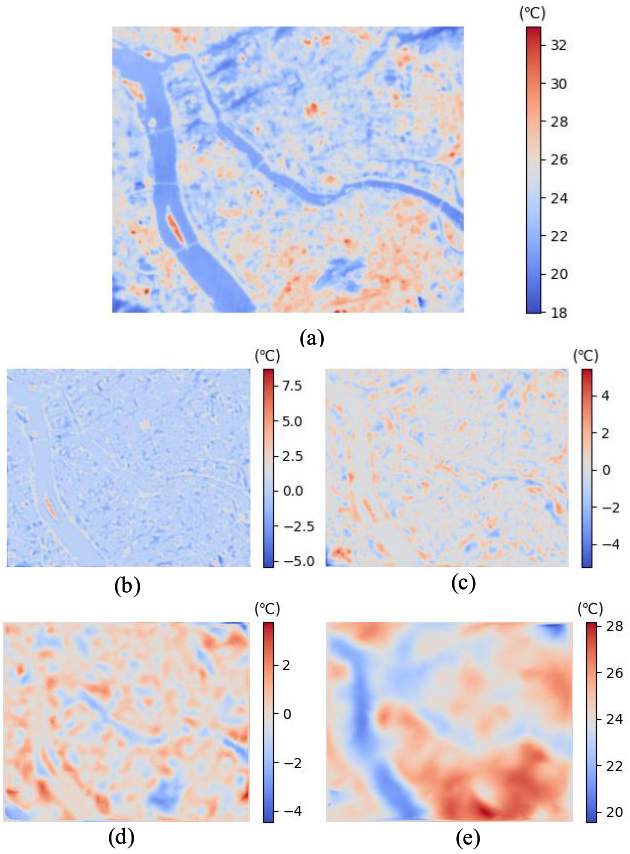


Fig. 5. Illustration of the result of BEMD, each colorbar's number represents the temperature in Celsius (°). (a) LST of Fuzhou, China. (b)–(d) IMFs that reflecting the temperature signals in different level of detail information. (e) Residual term from BEMD.

B. Network Architecture

The proposed PIHP-Net, as shown in Fig. 6, consists of two parts: multiscales encoder and information parallel decoder.

1) Multiscales Encoder: For the sake of multiscale GSCFM data, accordingly we apply the multiscale encoder to them. The input of our multiscales encoder is varied spatial scale GSCFM (control by different M), NDVI, NDBI, (match to the GSCFM) and forcing. Here we set $M = 35 + 10 * i, i = 1, \ldots, n$ to obtain different scales GSCFM, then leveraging local-surface feature extractor to encode the feature from them, respectively.

After the encoding operation, the local surface feature information into latent vector F_i, for $i = 1, \ldots, n$. All F_i are then concatenated, forming a latent feature map P, with size of $1024 \times n$. Next, the multilayer perceptron (MLP) is applied to integrate the latent vector P into the final latent vector F_0, size of $F_0 = 1024$, which contains different scales of GSCFM information. A batch of independent Forcing feature extractor will be used to extract the large-scale atmospheric feature G_i, for $i = 0, \ldots, n$. For all independent channel, we concatenate F_i and G_i to generate H_i, which contains the information of broader-scale atmospheric forcing factors and high-resolution local urban surface feature information.

The details of local-surface feature extractor and forcing feature extractor are as follows.

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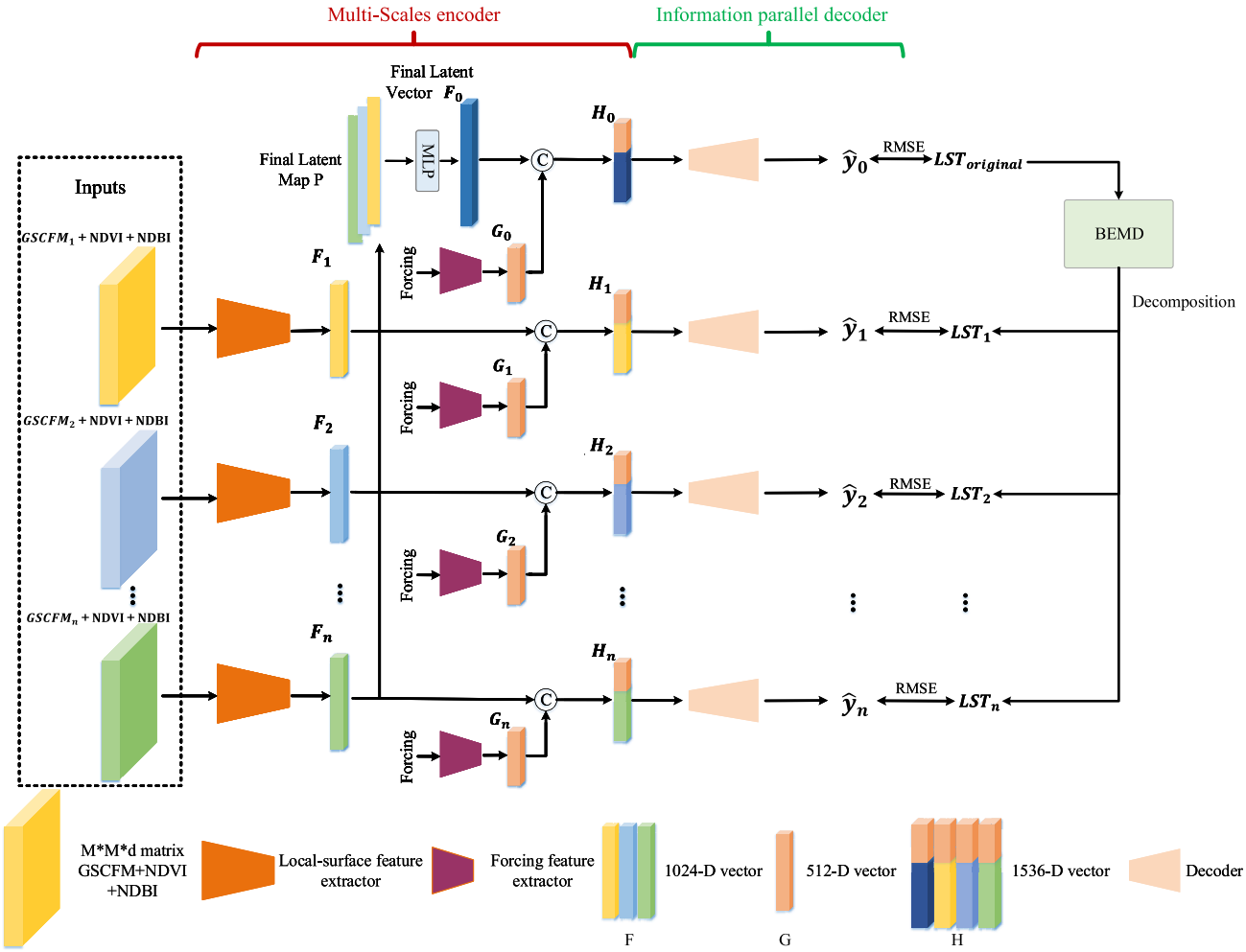


Fig. 6. Overview of the proposed physics-informed hierarchical perception network (PIHP-Net).

a) Local-surface feature extractor: To dig the structural information of the urban surface, we first introduce ResNet [39], which aims to extract local surface features and spatial structure information from the proposed GSCFM. This branch enables the neural network to capture high-resolution variations in the temperature of the urban surface. This network consists of a five-stage deep residual network, each stage containing two residual blocks. The first block is composed by two 3×3 convolutional layers sequentially. The stride is set as 2 and 1, respectively, and a skip connection is used to align the output shapes. The second block consists of two convolutional layers, where the stride is set as 1. The last stage consists of a 3×3 convolution operator, which adjusts the output size instead of a pooling operation. Batch normalization and Rectified Linear Unit (ReLU) layers are applied after each convolutional layer.

b) Forcing feature extractor: Considering the impacts of atmospheric states on the LST, we employ a MLP to encode the primary atmospheric forcing variables from the physics-based climate model. This procedure can be considered as a deep learning "solver" of the physical equations in those process-based models, in a way to mimic the dynamic simulations [10]. For the MLP, we introduce five layers to encode the feature vector into a vector G, with size 256. Scaled exponential Linear Units (SeLU) [49] is used to avoid gradient explosion and vanishing.

2) Information Parallel Decoder: To decode the aggregated feature H_i to predict LST, we fed them into regression branch, respectively. Each branch is comprised of three fully connected layers. Next, each H_i is mapped to temperature $\hat{Y}_i$ by the corresponding regression branch. Following the previous work [36], we employ the root mean-squared error (RMSE) and the L_2 normalization of the network weights to measure the loss.

3) Forcing Correction: In the whole framework, the upper-level atmosphere condition is related to the forcing data, in which humidity, radiation, precipitation, and the temperature of the upper atmosphere are involved. However, due to the fact that the LST is actually related to the near-surface temperature, there is an intrinsic deviation between the mean value of LST and the forcing data. To demonstrate this, we collect about 40 pieces of data over different seasons for 30 major cities under different climate regions in China. Then the mean value of the underlying surface in the LST data, along with mean value of upper air in the corresponding forcing data are applied to generate a scatter plot, as shown in Fig. 7.

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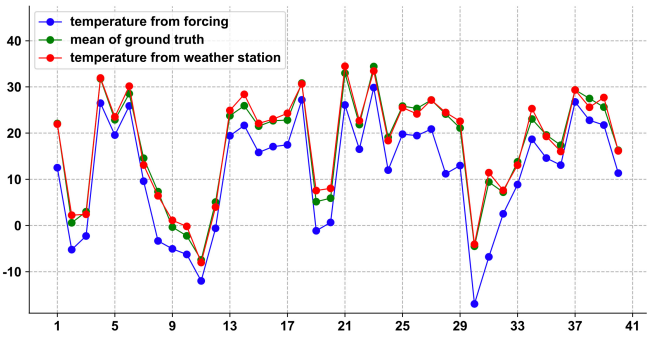


Fig. 7. Comparing the temperature from weather stations, forcing and average LST. The ordinate represents the temperature in Celsius (°C). The abscissa represents the city number.

In Fig. 7, the average 6.38 K distance between the LST and forcing can be observed, and such bias may largely affect the prediction result. To address this issue, we initially planned to find the mapping relationship between the mean value of LST and forcing by a simple branch in the network. However, limited by the period of satellite, the LST data are sparse and thus difficult to train the branch. Therefore, in our approach, the data from the weather station are included as the intermediate. The weather station data are dense and easy to acquire for cities, and most importantly, it matches really well with the mean value of LST. As shown in the red line in Fig. 7, the average bias of 0.42 K to the LST can be observed. Such an observation inspires us to fit the mean value of weather station by the input forcing, thus to correct the upper air forcing closer to the near-surface cases.

Specifically, the branch is simple: the mean value of LST is predicted by leveraging the MLP to fit the weather station data, and then the middle result is used as the input of the decoder network.

4) Loss Function: Following the previous work [36], we employ the RMSE and the L_2 normalization of the network weights to measure the loss. Each H_i, for $i = 0, \ldots, n$, has independent loss $Loss_i$, which is used to guide the training of the network, and the final loss L is the sum of them. The overall loss can be written as

$$L = \sum_{i=0}^{n} \text{Loss}_i \tag{4}$$

$$\text{Loss}_i = \underset{\mathbf{W},b}{\text{argmin}} L(\text{LST}_i, \hat{Y}_i) + \lambda R(\mathbf{W}) \tag{5}$$

$$L(Y, \hat{Y}) = \frac{1}{N} \sum_{j=1}^{N} \left(y_j - \hat{y}_j\right) \tag{6}$$

$$R(\mathbf{W}) = \|\mathbf{W}\|_2. \tag{7}$$

Here, Y_i, for $i = 1, \ldots, n-1$ is LST_i, which is generated by BEMD, and Y_0, Y_n represent the ground truth set. $\hat{Y}_i$, for $i = 0, \ldots, n$ represent the predicted results set from H_i. $\mathbf{W}$ and b are the combined coefficients of weights and bias terms, $y_j \in Y, \hat{y}_j \in \hat{Y}$, $j = 1, \ldots, N$, N is the data size of $Y, \hat{Y}$, and λ is the weight of the regularization term.

IV. Results and Discussion

A. Implementation Details and Baselines

1) Implementation Details: The models are evaluated quantitatively on test sets based on the root mean square error (RMSE), defined as

$$\text{RMSE} = \sqrt{\sum_{i=1}^{n} (\hat{y}_i - y_i)^2 / n}. \tag{8}$$

We implement our PIHP-Net on PyTorch [50]. The network uses end-to-end training mode. In the training phase, we adopt Adam solver [51] with an initial learning rate 0.001, which decayed by $0.5^{1/500}$ for each epoch.

2) Baselines: To our knowledge, the proposed PIHP-Net is the first physics informed deep neural network, which attempt to estimate the urban temperature at an high resolution (60-by-60 meters). In consideration, the huge calculated amount of the dynamic equation-based methods at such resolution, our comparisons are mainly focused on previous statistic model-based methods, such as linear regression [52], K-nearest neighbors (KNNs) regression [53], random forest (RF) regression [54], and deep neural network-based method, such as ResNet [39]. All the statistic model-based methods are implemented based on Scikit-learn [55], specifically, the parameter settings of these methods are as follows:

1) **Linear Reg** uses linear regression for simulation. We concatenate GSCFM and forcing as input. Specifically, each dimension of the $M \times M \times d$ matrices is averaged to a specific value, and the original matrix is reshaped to a $1 \times d$ vector. This vector is then imported into various regression methods. That is, linear models take the form

$$y = \sum_i \beta_i x_i + \varepsilon \tag{9}$$

where y is the response (LST), x_i is GSCFM and forcing reshaped to a $1 \times d$ vector, β_i is how the LST changes linearly with each x_i, and ε is the normally distributed error.

2) **KNN Reg** is KNN regression. Specifically, the number of neighbors in the KNN regression is set as 4. The maximum depth of the tree is set as 30. The input feature is the same as Linear Reg.
3) **RF Reg** is RF regression. The number of trees in the RFs regression is set as 150.
4) **ResNet** is based on ResNet50, and the corresponding parameters are the same as the ResNet in our network.

B. Ablation Study and Sensitive Analysis

In this section, we design a series of experiments to evaluate the effects of the various components in the proposed PIHP-Net.

1) Ablation of Hierarchical Urban Surface Perception Scheme: The motivation of the proposed BEMD-based hierarchical perception scheme is that PIHP-Net senses the surface structure information at different scales. The hierarchical perception scheme helps network more accurately simulate the

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TABLE II

AVERAGE TEMPERATURE RMSE OF ABLATION OF HIERARCHICAL URBAN SURFACE PERCEPTION SCHEME

	Guangzhou	Zhangzhou	Shenyang	Avg. Error (Kelvin)
PIHP-Net^0	0.62	0.64	0.66	0.64
PIHP-Net^1	0.58	0.58	0.62	0.59
PIHP-Net^2	0.56	0.56	0.59	0.57
PIHP-Net^3	**0.52**	**0.53**	**0.55**	**0.53**
PIHP-Net^4	0.56	0.59	0.63	0.59

TABLE III

RMSE FOR DIFFERENT LAND COVERS OF PIHP-NET^3 IN THE ABLATION OF HIERARCHICAL URBAN SURFACE PERCEPTION SCHEME

	Guangzhou	Zhangzhou	Shenyang	Avg. Error (Kelvin)
water	0.32	0.36	0.39	0.36
soil	0.41	0.45	0.49	0.45
roads/pavements	0.58	0.61	0.54	0.58
buildings	0.62	0.63	0.60	0.62
vegetation	0.45	0.49	0.51	0.54
unknown	0.71	0.66	0.75	0.71
total	0.52	0.53	0.55	0.53

TABLE IV

ABLATION OF HOW THE AGGREGATE BRANCH AFFECT THE RESULTS

	Guangzhou	Zhangzhou	Shenyang	Avg. Error(Kelvin)
PIHP^3 without aggregate branch	0.62	0.65	0.70	0.66
PIHP^3 with aggregate branch	**0.52**	**0.53**	**0.55**	**0.53**

TABLE V

AVERAGE TEMPERATURE RMSE OF SENSITIVE ANALYSIS OF PIHP NETWORK FOR INPUT DATA ERROR

	Guangzhou	Zhangzhou	Shenyang	Avg. Error (Kelvin)
without error	0.52	0.53	0.55	0.53
5% error of labeled remote sensing image	0.56	0.58	0.61	0.58
10% error of labeled remote sensing image	0.59	0.61	0.64	0.61
15% error of labeled remote sensing image	0.62	0.65	0.70	0.66
10% error of LST	0.59	0.61	0.61	0.60
20% error of LST	0.65	0.63	0.71	0.66
30% error of LST	0.71	0.78	0.82	0.77

surface temperature of the city. A group of experiments is designed to demonstrate the effectiveness of such a scheme, and confirm the appropriate layer number.

We selected 12 pieces of data, for the cities Guangzhou, Zhangzhou, and Shenyang in China, between January 1, 2015 and December 31, 2019. These data are selected to cover different seasons of the year, and to ensure the reliability of the evaluation results. All the data are handled as cloud-free to avoid aberrant temperature sampling by satellites. For each piece of data, the 70% area is chosen as the training data, and the rest 30% is applied for testing.

In the experiment, the average RMSE of the data for the PIHP-Net with different numbers of the layer are collected, and the corresponding results are shown in Table II. Where PIHP-Net$^\circ$ denotes the network without hierarchy, i.e., with number of branch 1. PIHP-Net^1, PIHP-Net^2, PIHP-Net^3 and PIHP-Net^4 correspond to the number of the perception branch 2–5. The results demonstrate that the hierarchical perception scheme brings about 6% performance increasing (the average temperature RMSE decreased from 0.64 to 0.59). With the increasing number of perception branch, the best performance appears at PIHP-Net^3, corresponding to four branches, and the RMSE increased 17% in total. Such results demonstrate that the proposed hierarchical perception scheme makes the PIHP-Net better capture the nonlinear relationship between the LST and the urban surface structures. In addition, the average RMSE of the data for the PIHP-Net^3 with different land covers are collected, and the corresponding results are shown in Table III. Except for unknown, the average RMSE of roads/pavements and buildings are higher at 0.62 and 0.58 K, respectively. The average RMSE of water is the lowest, is 0.36 K. This difference may be due to the fact that the temperature of water varies less, while the temperature of roads/pavements and buildings are more sensitive to environmental as well as human factors.

2) Ablation of the Aggregate Layer: As shown in Fig. 6, in our proposed PIHP-Net, an aggregate branch of the original LST data is also involved. In the experiments, we observe that the original LST image contributes significantly to the prediction results. To verify this, an ablation study is designed. Whereas the experiment data and process are the same as Section IV-B1, the number of hierarchical layers is set as four, and the results with and without the aggregate branch are collected, as listed in Table IV. The average RMSE of various cities is shown separately; the overall performance gains approximately 19% increasing for different cities. The underlying reason of this improvement is likely that the aggregate branch can correct the error in the LST edge part of the decomposition by BEMD.

3) Sensitive Analysis of PIHP Network: In our proposed PIHP-Net, the input training data mainly consists of GSCFM constructed from the labeled remote sensing images as well as LST ground truth. To analyze the sensitivity of the network to input data errors, the sensitive analysis to them is conducted as follows:

The sensitive analysis experiments are based on the ablation studies in Table II (where the same data and cities are selected). We have added the noise to 5%, 10%, and 15% of the real data by random sampling. Where the noise has been added to two sources of data (label of remote sensing images and the LST ground truth) separately. For the LST data, where the noise of ± 2 K has been added to the clean data. The results of different percentage errors are collected, as listed in Table V.

For the labeled remote sensing images, the RMSEs under 5%, 10%, and 15% percent of noise, increase 9%, 15%, and 23%, respectively. For the LST data, the RMSEs are 13%, 24%, and 44% higher, respectively.

The supplemented experiments show that the error of the input training data has a significant impact on the experimental results, and the results are more sensitive to the errors of the LST than the label of remote sensing images.

C. Comparisons

To comprehensively compare the proposed approach and previous statistical models and deep neural network-based

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TABLE VI

OVERALL PERFORMANCE COMPARISON OF DIFFERENT APPROACHES ON THE EXPERIMENT OF SINGLE CITY FUTURE TEMPERATURE PREDICTION. A SMALLER VALUE INDICATES A BETTER PERFORMANCE. THE AVERAGE ERRORS OVER ALL THE NINE CITIES OF EACH METHOD ARE SHOWN IN THE LAST COLUMN

Model	RMSE (Kelvin)									
	Xiamen	Zhangzhou	Guangzhou	Hefei	Zhengzhou	Shijiazhuang	Yinchuan	Shenyang	Chongqing	Avg. Error
Linear Reg	4.61	4.26	5.05	4.50	5.63	4.44	6.23	3.37	4.13	4.69
KNN Reg	3.99	3.78	3.33	3.41	4.14	3.59	5.20	4.50	3.98	3.99
RF Reg	3.84	3.40	3.37	3.48	4.07	3.36	4.93	3.79	3.54	3.75
ResNet	2.72	2.18	1.75	1.81	2.63	2.32	3.25	2.10	2.47	2.36
PIHP-Net^0	2.60	2.03	1.65	1.79	2.57	2.20	3.06	1.96	2.34	2.24
PIHP-Net	**2.00**	**1.78**	**1.32**	**1.44**	**2.13**	**1.78**	**2.52**	**1.66**	**1.87**	**1.83**

methods, two additional groups of experiments are designed. Specifically, the first group of experiments focuses on the single city evaluation to verify the performance for estimating a certain city at different times. The second group of experiments, on the other hand, pays more attention to the large-scale evaluation, where most of the major cities in China are involved is showing the fidelity of the proposed PIHP-Net.

1) Single City Temperature Prediction: We conducted this experiment in nine key cities distributed in major regions of China. Data from a single city at multiple time points are applied for training, and the temperatures at other time points are applied for testing. Because the surface structure of a city barely changes in 1–2 years, while the atmospheric condition changes dramatically and is closely related to the seasons. Thus, for each city, 6–8 pieces of data over four seasons are chosen for training, and 2–4 pieces of data are applied for testing.

Methods of linear regression, KNN regression, and RF regression (denoted as Linear Reg, KNN Reg and RF Reg separately) and the previous deep network ResNet are employed for comparison. The proposed GSCFM is not included for these baselines because we aim to figure out how the physics-informed GSCFM affects the results. The corresponding parameter settings are mentioned in Section IV-A2. The comparison without the hierarchical perception scheme is also involved, denoted as PIHP-Net°.

The average RMSE for each city is collected, with corresponding results listed in Table VI and the average errors over all nine cities for each method are shown in the last column. It is viewed that, previous statistic model-based approaches get the average error more than 3.3 K over all the cities. The linear model got the worst results with 4.69 K, because the relationship between the LST, underlying surface, and upper atmosphere is complex and nonlinear. The results of KNN and RF regression are similar, and appear unstable over different cities (with about 3.33 K for the best case and 5.20 K for the worst case). Thanks to the proposed GSCFM and the hierarchical perception scheme, our approach obtains an average error of 1.83 K for all cases. It is worth noting that for some of the cities such as Zhangzhou, Guangzhou and Shijiazhuang, the prediction error is close to the typical observational error of the satellite (e.g., about 1 K). The average error of ResNet (i.e., without the physics information GSCFM and hierarchical perception scheme), is 2.36 K, i.e., 22% higher than our approach. The average error of PIHP-Net° is 2.24 K, i.e., 18% higher. A visualization of the results for the nine cities is shown in Figs. 8 and 9.

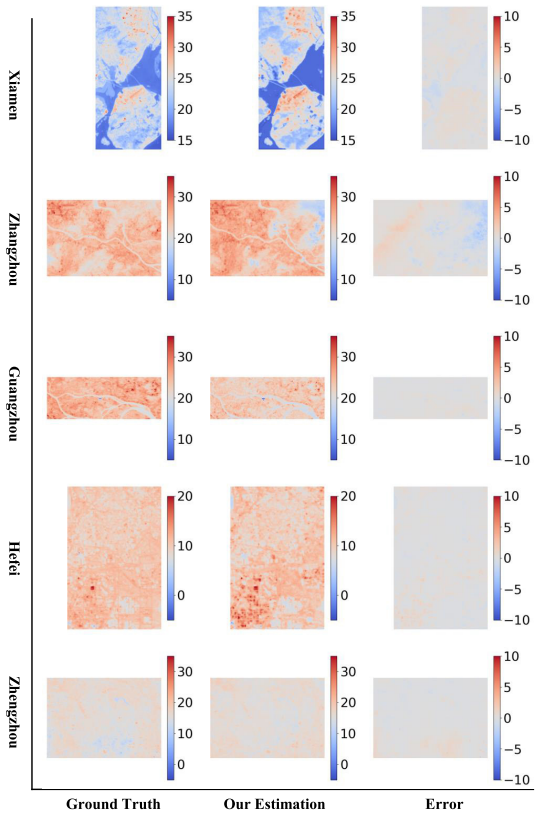


Fig. 8. Experiment of single-city temperature prediction visualization plots on Xiamen, Zhangzhou, Guangzhou, Hefei and Zhengzhou, China. Each colorbar's number represents the temperature in Celsius (°C). The vertical coordinates represent the city names and the horizontal coordinates represent the truth value, our simulation, and the error, respectively. All visualization plots are averaged results from the corresponding multiple data.

2) "Unseen" City Temperature Prediction: In this part, we attempt to evaluate the ability of PIHP-Net to predict the temperature of an unseen city in the training set. Such an experiment includes almost all the major cities in China. Considering the large area of China, this group of experiments is developed following the official major regions division of China, which are South, Southwest, Northwest, Northeast, Central, North, and East.

For each region, 2–4 cities, and four pieces of data for each city of over different seasons are chosen for training, as shown in Table VII. Then, the remaining 1–2 cities with 3–6 pieces of

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TABLE VII

SPLIT OF TESTING AND TRAINING CITIES FOR EACH REGION

Region	City for training	City for testing
South China	Nanning,Guangzhou,Haikou,Hong Kong	Macao
Southwest China	Guiyang,Chongqing,Chengdu	Lhasa
Northwest China	Xining,Yinchuan,Urumqi	Xi'an,Lanzhou
Northeast China	Changchun,Harbin	Shenyang
Central China	Wuhan,Changsha	Zhengzhou
North China	Shijiazhuang,Hohhot	Taiyuan
East China	Nanchang,Nanjing,Jinan	Hangzhou,Fuzhou

TABLE VIII

COMPARISON OF RMSE ON SEVEN REGIONS FOR TEMPERATURE PREDICTION OF TESTING CITIES USING PIHP-NET AND OTHER BASELINE METHODS. THE AVERAGE ERRORS OVER ALL TESTING CITIES OF EACH METHOD ARE SHOWN IN THE LAST COLUMN

Model	RMSE (Kelvin)									
	South China	Southwest China	Northwest China		Northeast China	Central China	North China	East China		Avg. Error
	Macao	Lhasa	Xi'an	Lanzhou	Shenyang	Zhengzhou	Taiyuan	Hangzhou	Fuzhou	
Linear Reg	5.06	7.03	6.84	6.30	5.22	5.70	3.49	5.66	6.83	5.7
KNN Reg	4.00	5.93	5.55	4.90	3.72	4.54	2.82	4.64	5.46	4.62
RF Reg	4.13	5.88	5.28	5.19	3.63	4.26	3.33	4.33	5.24	4.58
ResNet	2.20	4.44	4.20	3.66	2.03	2.52	1.74	2.44	2.60	2.87
PIHP-Net^0	1.91	3.93	3.76	3.42	1.83	2.25	1.54	2.23	2.42	2.59
PIHP-Net	**1.80**	**3.77**	**3.46**	**3.17**	**1.72**	**2.10**	**1.43**	**2.09**	**2.26**	**2.42**

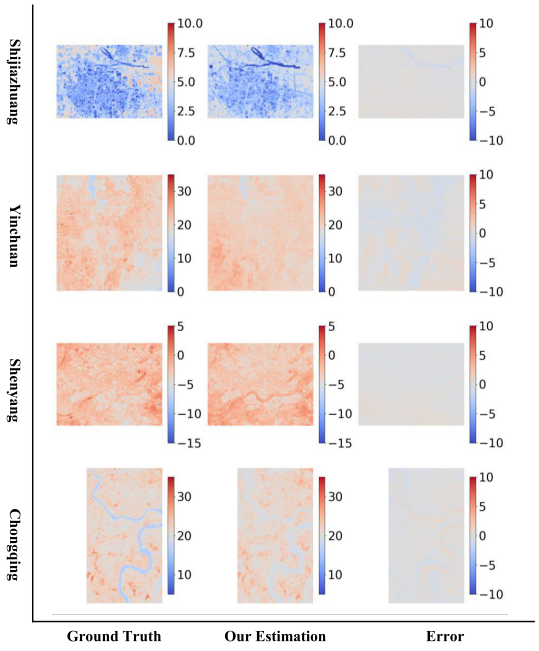


Fig. 9. Experimental single-city temperature prediction visualization plots on Shijiazhuang, Yinchuan, Shenyang and Chongqing.

data are applied for testing. Similarly, methods of Linear Reg, KNN Reg, RF Reg, ResNet and PIHP-Net° are employed for comparison. The average errors over all testing cities of each method are collected, as shown in Table VIII. The results show that, as the testing scale increases, the overall prediction error suffers, decreasing in different degrees for various methods. Specifically, results of previous statistical-based works remain unstable, maximum prediction error is more than 7.03 K at the city of Lhasa, and the average error over all the test cities also exceeds 4 K.

For the deep networks, average errors of 2.87, 2.59, and 2.42 K for ResNet, PIHP-Net° and our approach, respectively,

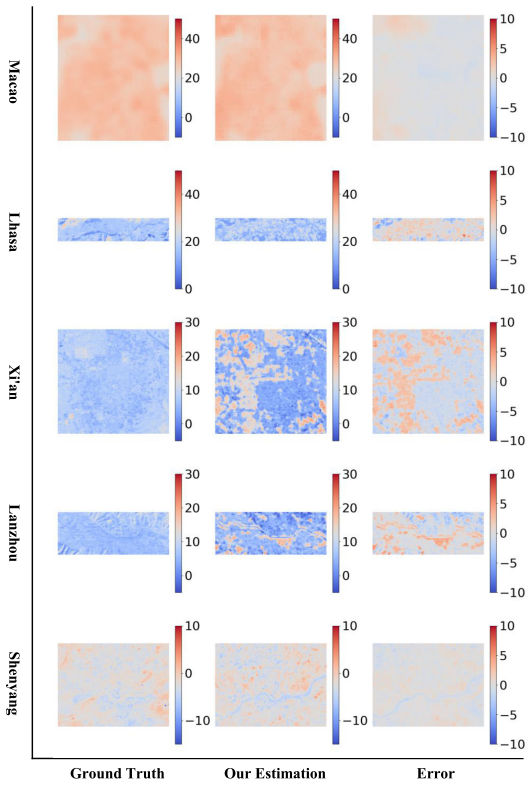


Fig. 10. Experimental "Unseen" city temperature prediction visualization plots on Macao, Lhasa, Xi'an, Lanzhou and Shenyang.

are observed. As shown in Figs. 10 and 11, due to the more complicated underlying urban surface and the vagaries of climate, the advantage of the proposed PIHP is still demonstrated but not as substantial as compared to the single city prediction.

D. Discussion

In this section, we focus mainly on the error analysis of the experimental results, where some of the interesting

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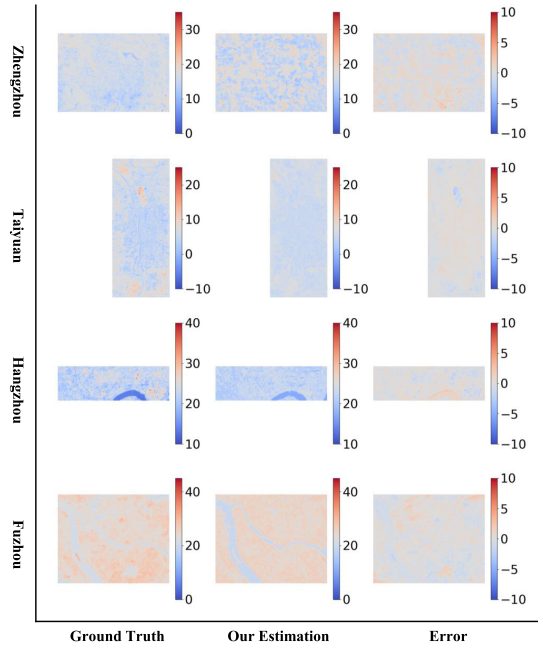


Fig. 11. "Unseen" city temperature experiment prediction visualization plots on Zhengzhou, Taiyuan, Hangzhou and Fuzhou.



Fig. 12. Seven regions of China. The red dots represent cities in the northwest and the yellow dots represent cities in the southwest.

factors besides the underlying surface and atmospheric forcing conditions are observed to react on the intracity LST. Thus, in this section, some of the instructive experimental results are presented, aiming to inspire more accurate intracity LST estimation systems in the future.

1) Errors From Regional Climate: As shown in Fig. 10 and Table VIII, it is worth to noting that, for two regions, Southwest and Northwest China, the prediction error for cities Lhasa, Xi'an, and Lanzhou significantly exceed the average. To find out the potential reasons, we first view the administrative division of China, as shown in Fig. 12. The Southwest and Northwest cover the largest area, about 50° longitude and 39° latitude, significantly larger than other regions. Such a large geographical span brings vagaries of climates for these two areas.

TABLE IX
RESULTS FOR THE FIVE CITIES ARE IN THE SPECIAL TRAINING SET

	Chengdu	Lanzhou	Xi'an	Urumqi	Xining
RMSE	1.88	3.42	3.38	3.71	4.03

For the Southwest, the testing city Lhasa is located on the Tibet Plateau, with an average altitude of 3650 m, while the training cities of this area Guiyang, Chengdu, and Chongqing, are all located at the Szechwan Basin, with an altitude of 250 m. Such geographical conditions bring extremely differential climate features for the training and testing cities, thus leading to the poor estimation performance.

For the Northwest, a similar situation is observed. Where Urumqi locates near the Junggar Basin, with desert all around. Xining locates on the Qinghai plateau, with an average altitude of 2200 m. Yichuan locates in the Helan mountain area. Lanzhou locates at Yellow River Valley basin, on the Loess Plateau. Xi'an locate at Kuan-chung Plain, with eight river systems around. All the five major cities in Northwest China have their own unique geographical and climatic characteristics, leading to biased estimations.

To verify such hypothesis, two additional experiments are applied specifically to these two regions. For the Southwest, Guiyang and Chongqing are applied for training to test Chengdu. For the Northwest, we plan to augment the network knowledge, so just one testing city is applied, here Lanzhou, Xi'an, Urumqi and Xining are applied as the testing cities separately. Corresponding results are listed in Table IX. The three cities in Southwest are with similar geographical and climatic conditions, thus getting satisfactory results 1.88 K. However, for the Northwest, due to the complex geography environment, the testing results are still not satisfactory (3.42 K for Lanzhou, 3.38 K for Xi'an, 3.71 K for Urumqi, and 4.03 K for Xining). This suggests how to introduce the unique local climatic factors into the network is an interesting point for future research.

2) Errors From Artificial Terrain: It is also worth noting that the artificial flat structures' large area often leads to higher errors, such as ports and airports. Fig. 13(b) and (d), correspond to airports and ports in Xiamen. The dates of LST from left to right in (c) and (e) are captured on March 11, 2018, February 25, 2019, and March 30, 2019, respectively. The climate conditions of these data are similar; however, the temperature represents high dynamic fluctuation. A possible reason could be that, for the flat structures, due to their characteristics of fast heat absorption and dissipation, the temperature tends to change rapidly even at different times of the same day. Such a property makes the urban LST highly correlate to the exact time point of the day when the data are captured; thus leading to higher prediction error at these areas.

3) Errors From Human Behavior: Errors related to anthropogenic activities can be observed from our results as well; such errors always occurred in the industrial estates regions.

One typical case is shown in Fig. 14(a) and (b), in the single-city temperature prediction experiment IV-C1,

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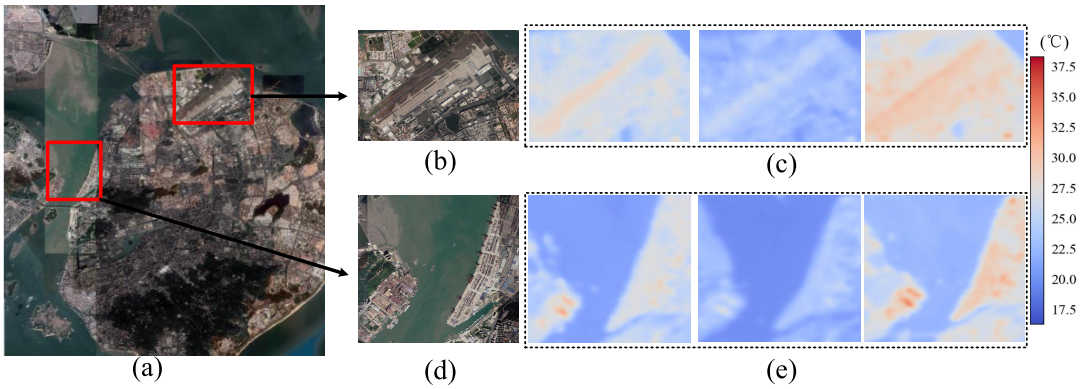


Fig. 13. Illustration of satellite images and LST in Xiamen. (a) Satellite remote sensing images of Xiamen. (b)–(c) Satellite image of Xiamen Gaoqi Airport and the LST data of the corresponding area. (d)–(e) Satellite images of Xiamen Haitian Pier and LST data of the corresponding area.

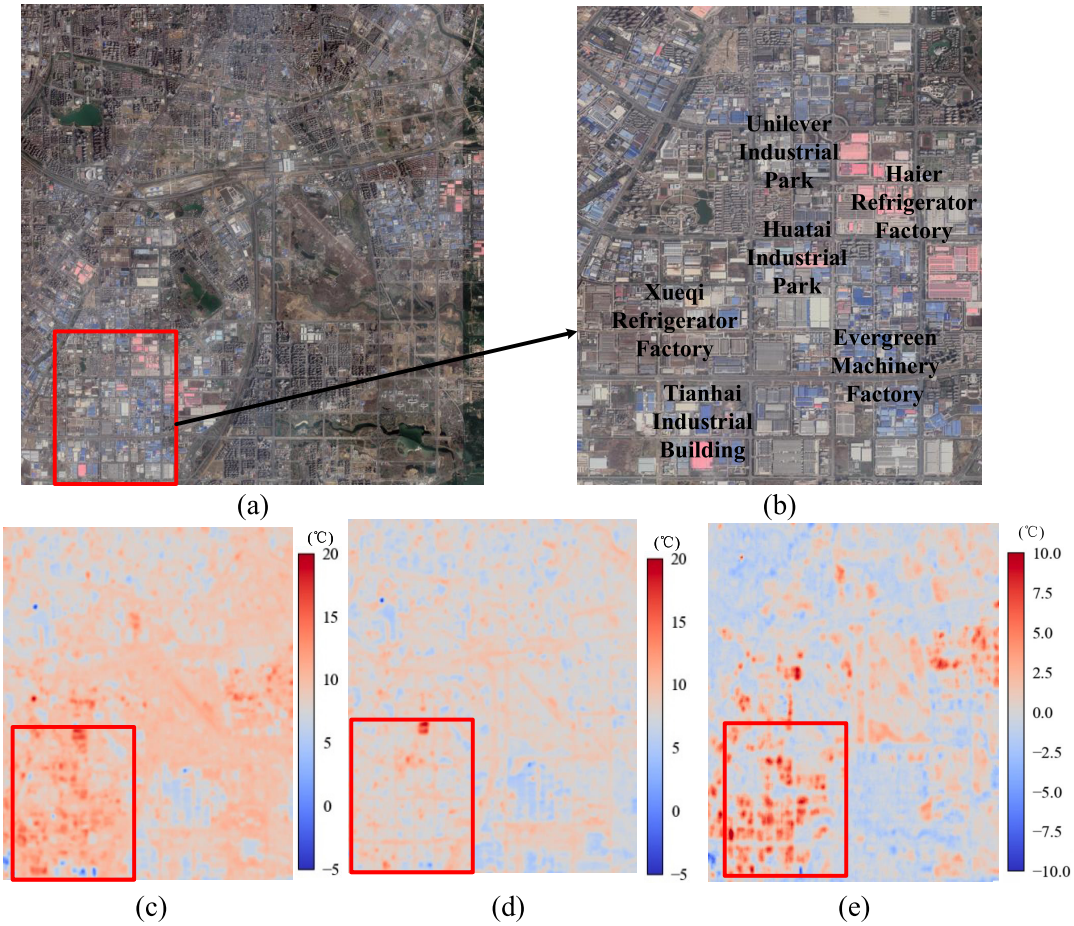


Fig. 14. Illustration of the prediction accuracy analysis of the experimental results in Hefei. (a) Satellite remote sensing images of Hefei. (b) Red boxed area in satellite map. (a) Contains a large number of factories. (c) LST visualization results for Hefei on January 23, 2019. (d) LST visualization results for Hefei on February 05, 2018. (e) Prediction error visualization of Hefei.

at Hefei city. The region highlighted by the red box in (a) contains a large number of factories and industrial campuses, and the prediction error in this region is shown in Fig. 14 (e), where dense high error points are observed.

The original LST data are visualized for further in-depth error analysis. Fig. (c) shows a typical training case, where a region with a local high temperature is observed in the red box area. Fig. (d) shows one of two applied testing samples, correspond to the dates of February 5, 2018. The high-temperature region that should have appeared has become less obvious. This is because the date of February 5, 2018 is near to the Chinese New Year holiday, and most of these

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factories are shut down, leading to higher errors in this region. Therefore, anthropogenic heat could have large impacts on the urban surface temperatures at such a high resolution, and such a factor should be considered in the future work of intracity temperature prediction.

V. Conclusion

In this article, we propose a PIHP-Net for high-resolution and high-precision urban surface temperature prediction from upper-atmospheric forcing fields and land surface satellite imagery. This network can be used to predict intracity urban LST in future time periods when the future forcing meteorology can be easily obtained from regional climate models. The PIHP-Net employees a designed multiscale encoder to construct different scales GSCFM, thus extracting different scales of urban surface structure. Based on this, a hierarchical urban surface perception scheme is proposed via a multibranch network structure. Such a scheme is able to capture the complex nonlinear relationship between the land surface and temperature in different scales. Extensive experimental results show that the PIHP-Net consistently outperforms previous baselines on the high resolution urban LST prediction tasks.

We conducted designed experiments 31 major or provincial capital cities in China to evaluate the performance of the proposed PIHP-Net comprehensively. The estimation errors of the previous static model-based method for 60-by-60 meters grids is about 6 K, for most cases of our experiments, the errors are less than 2 K, thus making it possible to estimate or predict *intracity temperature* from multispectral satellite imagery. Compared to previous statistical models, the proposed approach has at least more than 30% performance improvement, either for single or multiple city predictions. These results demonstrate the validity and efficacy of our proposed approach.

Major contributions of this article include the following three aspects:

1) We propose to exploit high-resolution, largely available multispectral satellite images to tackle the challenging high-resolution urban temperature estimation problem.
2) We develop, for the first time, a novel PIHP-Net, for accurate, high-resolution and generalizable urban surface temperature estimation. Benefiting from a proposed novel semantic category histogram and a hierarchical perception scheme, the PIHP-Net can generate accurate urban LST estimation results (at least more than 30% improvement compared to previous statistical model-based works) at a high resolution (60-by-60 meters grids).
3) We build a novel group dataset related to the urban LST. Such datasets cover all the major cities (31 cities were used) in China, which will support comprehensive and large-scale experiments for further research at this field.

Our results also suggest that surface structural properties and human activities greatly influence on the surface temperature at a high resolution, both spatially and temporally. For cities that span a large area, forcing meteorology, as well as urban forms, are the main factors that determine the surface temperature variations.

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Donghang Wu received the B.S. degree from Fujian Normal University, Fuzhou, China, in 2020. He is currently pursuing the M.S. degree with the School of Informatics, Xiamen University, Xiamen, China.

His research interests include remote sensing image processing and deep learning.



Weiquan Liu (Member, IEEE) received the B.S. and M.S. degrees in applied mathematics from the College of Science, Jimei University, Xiamen, China, in 2013 and 2016, respectively, and the Ph.D. degree in computer science and technology from the School of Informatics, Xiamen University, Xiamen, in 2020.

He is currently a Postdoc with the Information and Communication Engineering Postdoctoral Research Station, Fujian Key Laboratory of Sensing and Computing for Smart Cities, School of Informatics, Xiamen University. His research interests include remote sensing, computer vision, machine learning, mobile laser scanning point cloud data processing, and augmented reality.



Bowen Fang received the B.S. degree from Peking University, Beijing, China, and the M.S. degree in environmental science from the Yale School of the Environment, New Haven, CT, USA. He is currently pursuing the Ph.D. degree with the Department of Civil and Environmental Engineering, University of Illinois at Urbana-Champaign, Urbana, IL, USA.

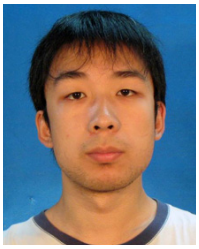
His research uses Earth system model and remote sensing to study the interactive impact of urbanization and climate change. He is also interested in urban heat island mitigation and other engineering solutions to sustainable urban development.

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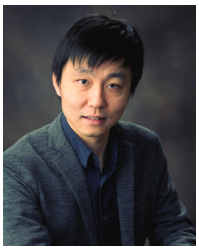
Linwei Chen received the B.S. degree from Fujian Normal University, Fuzhou, China, in 2019. He is currently pursuing the M.S. degree with the School of Informatics, Xiamen University, Xiamen, China.

His research interests include deep learning and its applications on 3-D point cloud processing.



Yu Zang (Member, IEEE) received the B.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 2008 and 2014, respectively.

He is currently a Research Associate Professor with the School of Informatics, Xiamen University, Xiamen, China. His main researches include remote sensing image processing, computer vision and graphics and mobile light detection and ranging (LiDAR) data analysis.



Lei Zhao received the B.S. degree in atmospheric physics from Nanjing University, Nanjing, China, in 2009, and the Ph.D. degree from Yale University, New Haven, CT, USA, in 2015.

He is currently an Assistant Professor with the Department of Civil and Environmental Engineering and an Assistant Professor affiliated with the National Center of Supercomputing Applications, University of Illinois at Urbana-Champaign, Urbana, IL, USA. Before joining the University of Illinois at Urbana-Champaign, he finished his postdoctoral training at Princeton University, Princeton, NJ, USA. His research interests include multiscale climate modeling, remote sensing, urban climate, environmental fluid mechanics ann turbulence, machine learning and statistical modeling. He has published more than 18 peer-reviewed articles in worldwide top-ranked journals including *Nature*, *Nature Climate Change*, *Nature Geoscience*, *Nature Communications* as first author and/or corresponding author.



Shenlong Wang (Member, IEEE) received the Ph.D. degree from the Department of Computer Science, University of Toronto, Toronto, ON, USA.

He worked as a Research Scientist working with Raquel Urtasun at the Uber Advanced Technologies Group (ATG), Pittsburgh, PA, USA. He is currently an Assistant Professor with the University of Illinois at Urbana-Champaign (UIUC) Department of Computer Science, affiliated with the Computer Vision at UIUC and Illinois Robotics Group. Before joining UIUC, he was a Visiting Scholar with Vladlen Koltun at the Intel Intelligent Systems Laboratory, Intel, USA. His research interests include 3-D computer vision and robotics, specifically on topics such as 3-D perception and modeling, photorealistic content creation, localization and mapping, robot simulation and self-driving.



Cheng Wang (Senior Member, IEEE) received the Ph.D. degree in signal and information processing from the National University of Defense Technology, Changsha, China, in 2002.

He is currently a Professor with the School of Informatics, and the Executive Director with the Fujian Key Laboratory of Sensing and Computing for Smart Cities, Xiamen University, Xiamen, China. He has coauthored more than 150 articles in referred journals and top conferences including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *Pattern Recognition*, the IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, the IEEE Conference on Computer Vision and Pattern Recognition, Association for the Advancement of Artificial Intelligence (AAAI), and International Society for Photogrammetry and Remote Sensing (ISPRS) the *Journal of Photogrammetry and Remote Sensing*. His research interests include point cloud analysis, multisensor fusion, mobile mapping, and geospatial big data.

Prof. Wang is a Fellow of the Institution of Engineering and Technology. He is also the Chair of the Working Group I/6 on Multisensor Integration and Fusion of the International Society of Remote Sensing.



José Marcato, Jr. (Member, IEEE) received the Ph.D. degree from São Paulo State University, São Paulo, Brazil, in 2014.

He is a Professor with the Faculty of Engineering, Architecture and Urbanism and Geography, Federal University of Mato Grosso do Sul (UFMS), Campo Grande, Brazil. He is the Head of the Geomatics Laboratory, UFMS and has experience in coordinating international projects. With expertise in photogrammetry and remote sensing, his research program is focused on unmanned aerial vehicle (UAV) and close-range photogrammetry, and deep learning in environmental and precision agriculture applications. He has published over 75 peer-reviewed journal articles, and has experience as an Associate Editor of two scientific journals and the Guest Editor of several special issues in refereed journals of remote sensing. He becomes in 2020 a Distinguished Researcher by the Brazilian National Council for Scientific and Technological Development.



Jonathan Li (Senior Member, IEEE) received the Ph.D. degree in geomatics engineering from the University of Cape Town, Cape Town, South Africa, in 2000.

He is currently a Professor with the Department of Geography and Environmental Management and cross-appointed with the Department of Systems Design Engineering, University of Waterloo, Waterloo, ON, Canada. His main research interests include image and point cloud analytics, mobile mapping, and artificial intelligence (AI)-based information extraction from LiDAR point clouds and earth observation images. He has coauthored over 500 publications, including more than 330 in refereed journals and more than 210 in conference proceedings.

Dr. Li is an Elected Fellow of the Canadian Academy of Engineering (CAE) and the Engineering Institute of Canada (EIC). He was a recipient of the 2021 Geomatica Award, the 2020 Samuel Gamble Award, and the 2019 Outstanding Achievement in Mobile Mapping Technology Award. He is currently serving as the Editor-in-Chief for the *International Journal of Applied Earth Observation and Geoinformation*, and an Associate Editor for IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and *Canadian Journal of Remote Sensing*.

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